

stockplotr: :CHEATSHEET

Create report-ready tables and figures for stock assessments.



The Basics

Convert output file to a standardized framework.

```
data <- convert_output(  
  file = <FILE_PATH> ,  
  model = "MODEL ACRONYM" ,  
  fleet_names = c("FLEET_NAMES") ,  
  save_dir = <FILE_PATH>  
)
```

Optional

label	estimate	year	fleet	unit	...
spawning_biomass	2017090	2016	NA	kg	...
spawning_biomass	2286740	2017	NA	kg	...
...	...	...	...	...	...

Create customizable plots.

```
plot_biomass(  
  dat = <DATA> ,  
  geom = <GEOM> ,  
  group = "<COLUMN_NAME>" ,  
  facet = "<COLUMN_NAME>" ,  
  ref_line = "msy" ,  
  unit_label = "metric tons" ,  
  module = "MODULE_NAME" ,  
  scale_amount = 1 ,  
  relative = FALSE ,  
  make_rda = FALSE ,  
  figures_dir = <FILE_PATH> ,  
  interactive = TRUE ,  
  ...  
)
```

"line", "point" or "area"

Export object as an .rda

Additional ggplot2 arguments

Example

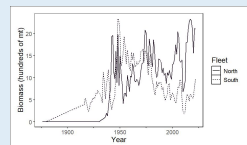
```
plot_biomass(example_data, unit_label = "mt", ref_line = "target",  
  scale_amount = 100, module = "TIME_SERIES",  
  make_rda = TRUE, figures_dir = getwd())  
load(biomass_figure.rda)
```

```
rda$figure  
rda$caption
```

```
> "Biomass (B) time series. The horizontal dashed line  
represents the limit reference point (msy mt)."
```

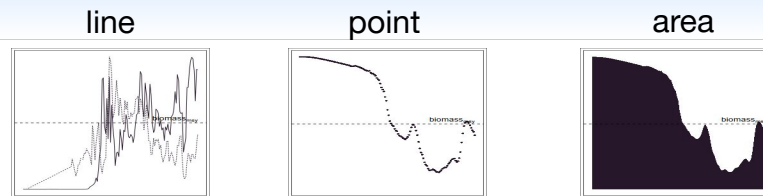
```
rda$alt_text
```

```
> "Line graph showing biomass time series. The x axis  
shows the year, which spans from 1874 to 2022. The y  
axis shows biomass in hundreds of mt, which spans from  
54.2 to 426.49."
```



Figures

time series



plot_biomass(dat, geom, ...)

Total biomass of line, point, or area over time with error where applicable.

plot_discard(dat, ...)

Observed vs. estimated discards by fleet over time.

plot_fishing_mortality(dat, ...)

Line or points of fishing mortality over time.

plot_landings(dat, ...)

Line, point, or area over time.

plot_natural_mortality(dat, ...)

Line graph of natural mortality over age classes.

plot_recruitment(dat, ...)

Observed and expected recruitment over time with error where applicable.

plot_selectivity(dat, ...)

Line graph of selectivity over age or length classes.

plot_spawning_biomass(dat, ...)

Line, point, or area over time with error where applicable.

plot_stock_recruitment(dat, ...)

Stock recruitment relationship with fitted line.

age series

plot_abundance_at_age(dat, z, proportional, ...)

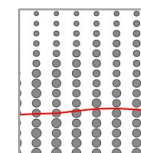
Bubble plot depicting abundance at age over time with trend lines across the x and y axes.

plot_biomass_at_age(dat, ...)

Bubble plot representing biomass at age over time.

plot_catch_comp(dat, ...)

Bubble plot representing catch at age over time with trend lines of top 5% strongest cohorts.



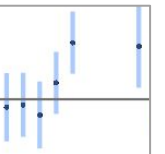
miscellaneous

plot_index(dat, ...)

Observed vs. estimated indices of abundance by fleet with error where applicable.

plot_recruitment_deviations(dat, ...)

Scatter plot of points with line segments depicting error.



Tables

Tables follow the same conventions of figures (except for facetting).

table_biomass_abundance_catch(dat, ...)

Total biomass, abundance, and catch/landings over time.

table_indices(dat, ...)

Indices of abundance by fleet over time.

table_landings(dat, ...)

Total or indexed landings over time including error.

table_projections(dat, ...)

Projected spawning biomass, fishing mortality, and catch.
Included in the {asar} report template Executive Summary.

table_total_catch(dat, ...)

Total catch over time including error.

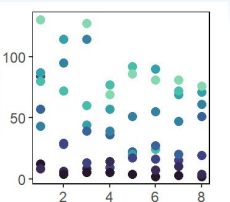
Theming

theme_noaa()

Apply NOAA Fisheries theme to ggplot2 objects using (+) operator.

theme_table(table)

Apply NOAA Fisheries theme to gt objects.



Exporting

save_all_plots(dat, ...)

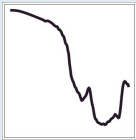
Save each available plot and table in stockplotr as an rda in a folder called "figures".

Follow along with our planning and development on our [GitHub project](#) and [milestones](#)! Be on the lookout for future releases including more tables and figures!

Figures

Plot Types

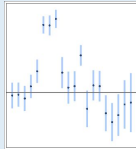
(All base plotting functions accept `dat` (data), `x`, `y`, `group`, `facet`, `xlab`, and `ylab`.)



plot_timeseries(dat, geom, ...)

Line, point, or area over time with error where applicable.

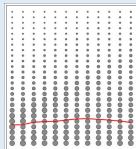
```
plot_timeseries(  
  dat = data, group = "fleet", facet = "area")
```



plot_error(dat, ...)

Scatter plot of points with line segments depicting error.

```
plot_error(petrade, ...)
```



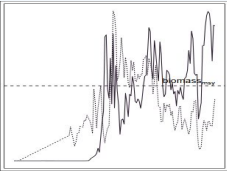
plot_aa(dat, z, proportional ...)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

```
plot_aa(petrade, facet = "area", proportional = TRUE)
```

Automated

Standard XY Plots time series



```
plot_spawning_biomass(  
  dat, geom, group, facet, ref_line, unit_label, scale_amount,  
  relative
```

```
plot_biomass(dat, ...)  
plot_recruitment(dat, ...)  
plot_landings(dat, ...)  
plot_fishing_mortality(dat, ...)
```

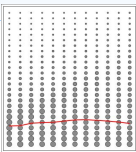
age series

```
plot_natural_mortality(dat, ...)
```

miscellaneous

```
plot_spawn_recruitment(dat, ...)
```

“At-age” Bubble Plots



plot_abundance_at_age(dat, z, proportional ...)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

plot_biomass_at_age(dat, ...)

Bubble plot representing biomass by age over time.

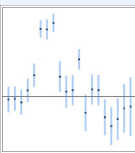
```
plot_biomass_at_age(petrade, facet = "sex")
```

plot_catch_comp(dat, ...)

Bubble plot representing catch by age over time with trend lines of top 5% strongest cohorts.

```
plot_catch_comp(petrade, ...)
```

Error



plot_recruitment_deviations(dat, ...)

Scatter plot of points with line segments depicting error.

```
plot_recruitment_deviations(petrade, ...)
```

Theming + Customization

```
a + theme_noaa(discrete = TRUE)
```

```
plot_spawning_biomass(  
  example_data,  
  group = "fleet") +  
ggplot2::geom_hline(ggplot2::aes(  
  yintercept = 1000))
```

Tables

Prepare data for table

prepare_table()

Filter data to be ready to input into a plotting or table function.

```
data <- stockplotr::prepare_table(  
  dat = data,  
  label_name = "quantity name to plot",  
  geom = "line",  
  era = NULL,  
  group = "indexing column",  
  facet = "column(s) for facetting",  
  interactive = TRUE  
)
```

Optional:
"point" or
"area"

Year	Biomass (mt)	Catch (mt)	Fishing Mortality
2000	2017090	43092	0.20
2001	2286740	42095	0.21
2002	2190210	90283	0.43
...	...	...	...

Plot Types

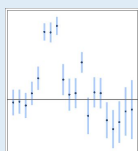
(All base plotting functions accept dat (data), x, y, group, facet, xlab, and ylab.)



plot_timeseries(dat, geom, ...)

Line, point, or area over time with error where applicable.

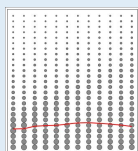
`plot_timeseries(petrале, group = "fleet", facet = "area")`



plot_error(dat, ...)

Scatter plot of points with line segments depicting error.

`plot_error(petrале, ...)`



plot_aa(dat, z, proportional ...)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

`plot_aa(petrале, facet = "area", proportional = TRUE)`

Automated Functions

Create tables – no work required.

```
table_spawning_biomass(geom = "line")  
dat, geom, group, facet, ref_line, unit_label, scale_amount,  
relative
```

```
table_spawning_biomass(geom = "point")  
dat, geom, group, facet, ref_line, unit_label, scale_amount,  
relative
```

```
table_spawning_biomass(geom = "area")  
dat, group, ref_line, unit_label, scale_amount, relative
```

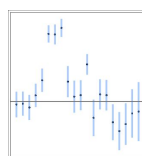
```
table_biomass(dat, ...)
```

```
table_recruitment(dat, ...)
```

```
table_spawn_recruitment(dat, ...)
```

```
table_landings(dat, ...)
```

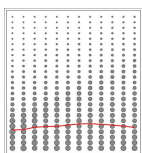
```
table_fishing_mortality(dat, ...)
```



plot_recruitment_deviations(dat, ...)

Scatter plot of points with line segments depicting error.

`plot_recruitment_deviations(petrале, ...)`



plot_abundance_at_age(dat, z, proportional ...)

Bubble plot depicting estimates at age over time with trend lines across the x and y.

plot_biomass_at_age(dat, ...)

Bubble plot representing biomass by age over time.

`plot_biomass_at_age(petrале, facet = "sex")`

plot_catch_comp(dat, ...)

Bubble plot representing catch by age over time with trend lines of top 5% strongest cohorts.

`plot_catch_comp(petrале, ...)`

Exporting

1. Execute all functions at once
2. Render and compile all functions into a readable html for interpretation

save_all_plots(dat, ...)

Save each available plot and table in stockplotr as an rda in a folder called "figures".

`save_all_plots(petrале, ...)`

html_all_figs_tables(dat, ...)

Create an html file containing all plots and tables from stockplotr to view together.

`save_all_plots(petrале, ...)`

Theming

Create and automate alternative text & captions. Any plot in stockplotr exported as an .rda will contain alternative text and a caption customized to the data using key quantities. A csv file is generated containing captions and alternative text associated with every plot and table that could be produced.

label	type	caption	alt_text
spawning_biomass	figures	Model-estimated spawning biomass...	Line graph showing ...
catch	table	Observed catch over time...	NA
recruitment	figure	Estimated recruitment by ...	Line graph showing...
...	...	...	...

.RDA Package & Key Quantities

- `make_rda = TRUE` → produce .rda file containing (1) figure/table, (2) caption, (3) alternative text
- Quantities extracted from data file
- Replace **placeholders** in caption and alternative text to maintain standardization while customizing reported values

```
plot_spawning_biomass(  
  dat = petrале,  
  make_rda = TRUE  
)  
load("figures/spawning.biomass_figure.rda")  
rda$alt_text  
rda$cap
```

asar’s former convert_output() section in cheatsheet

🏠→△ Convert model output

Convert output file to a standardized framework

```
output <- asar::convert_output(  
  output_file = "Report .sso",  
  model = "ss3")
```

SS3 files supported

```
output <- asar::convert_output(  
  output_file = "Report .rdat",  
  model = "bam")
```

BAM files supported

```
output <- asar::convert_output(  
  output_file = estimates,  
  model = "fims")
```

FIMS files supported
(tibble in R environment (shown here) OR .rds file)

Standardized framework format (dataframe in .rda file)

label	estimate	year	fleet	sex	...
spawning_biomass	2017090	2016	NA	NA	...
spawning_biomass	2286740	2017	NA	NA	
...	...	...	...	...	...

 See “Standardizing Assessment Model Output” vignette

